

GENERAL RISK ASSESSMENT TEMPLATE

Work area / operation	CB11.10.403 – Lab Assignment 2 pick-and-place workcell (UR3e + Stäubli TX2-60, optional PLC safety extension)		Assessor's name	Rudra Narayan Sahu	
Other persons consulted	Jayden Kim			Date of safety assessment	30/09/2026
Subject Coordinator's Name	Gavin Paul	Lab Supervisor's Name		Felipe Ulloa Gutierrez	

ACTIVITY - Describe hazardous activities related to the work area or operation.	ASSOCIATED HAZARDS	INHERENT RISK - Harm that could occur from these hazards if controls fail or are not in place.	EXISTING CONTROL MEASURES	PROPOSED CONTROL MEASURES - Proposed action to minimise risk to an acceptable level.	TARGET DATE - To implement proposed controls	RESIDUAL RISK LEVEL (H,M,L)
Controlling the real UR3e from Python/ROS (group GUI, jog/teach interface) during the pick-and-place task	Unexpected motion from code bugs, wrong IK solution, coordinate-frame or unit errors Stale/delayed ROS messages or network drop-out Teach pendant and Python both sending commands at once	Impact or crush injury to hands, arms or head Damage to robot, gripper and fixtures	UR3e cobot force/speed limits and protective stop Teach pendant e-stop Tutor supervision and lab induction	- Run every trajectory in the Swift simulation first using the same frames, joint limits and collision checks - First real runs at reduced speed (speed slider $\leq 30\%$) with conservative waypoints inside a known safe workspace - One clear command source at a time; operator announces "moving" before every command - Spotter keeps a hand near the pendant e-stop	Before first real run, then ongoing	L
Picking and placing objects with the gripper	Pinch points between gripper fingers, object and table, and between robot links Object dropped or thrown during fast motion Payload above rating	Finger pinch/crush injury Bruising or eye injury from a dropped or thrown object	Light, blunt test objects only (plastic/foam blocks) Gripper and fixtures approved by tutor Bins and fixtures fixed to the bench	- Keep objects well under the rated payload including gripper mass (UR3e 3 kg, Stäubli TX2-60 up to 4.5 kg) - Never reach into the gripper or between object and fixture while the robot is enabled; pause/stop first - Keep release heights low; test grip at low speed first	Ongoing	L

Operating the Stäubli TX2-60 (6-axis industrial robot with CS9 controller and SP2 teach pendant) in the same workcell as the UR3e (Simulated in this project – assessed as if installed on real hardware)	Industrial robot: much faster and stronger than the UR3e and not force-limited like a cobot unless its safety functions are set up Reach overlapping the UR3e workspace; robots colliding or trapping a hand Unexpected motion from wrong program, frame or speed settings, or when switching between manual and automatic modes Heavy arm (about 50 kg) and mains-powered CS9 controller	Serious impact, crush or pinch injury Collision damage to both robots, grippers and fixtures	CS9 SIL3-PL safety functions (Safe Speed, Safe Zone, Safe Stop, Safe Tool) SP2 pendant e-stop and enabling device Supervised use only	• Install inside a guarded cell (fencing with interlocked door or light curtain); use only with a supervisor present • Manual mode with reduced speed while teaching and testing; automatic mode only with everyone outside the cell and tutor approval • Set Safe Zone/Safe Speed limits and keep its workspace separate from the UR3e, or use a shared-zone lock in code • Pendant held by the operator with e-stop in reach; test the e-stop and enabling device every session • Test every motion in simulation first	Before first use, then ongoing	M
Group members, tutors and other students around the workcell during testing, the demonstration and video filming	People entering the robot workspace while it moves Distraction while filming or presenting Crowding around the bench	Impact injury to a person inside the workspace Trips and falls	Tutor supervision Lab rules from the CB11.10.403 induction	• Mark a keep-out zone with tape/barriers around both robots • Only the operator and spotter at the cell; everyone else outside the zone • Warn nearby people before motion; film from outside the zone • Stop straight away if anyone enters the zone or the light curtain is broken	Ongoing	L
Emergency stop, protective stop and restart/recovery	Robot resuming by itself after a stop Software (GUI) e-stop is not safety rated and may fail Old trajectory still queued after reset Restarting from an unknown pose	Unexpected motion on restart causing impact or crush injury	UR3e hardware e-stop and protective stop GUI e-stop button in the group's Python app	• Treat the GUI e-stop as a secondary stop only; the hardware e-stop is always within reach • After any stop: find the cause, deliberate reset/acknowledge, then a separate start/resume • Clear queued commands and re-read joint states before resuming • Recover to the home pose at reduced speed; tell the tutor about every stop	Ongoing	L

Using plug-in equipment: laptops, Raspberry Pi/ROS computer, UR3e and Stäubli CS9 controllers, power supplies, USB and Ethernet cables	Damaged or exposed cables Cables across walkways (trip hazard) Cables caught by a moving robot Mains-powered robot controllers	Electric shock or burns Trip and fall injury Cable snag pulling equipment off the bench	Lab equipment tested and tagged Cables routed away from walkways	• Inspect cables and plugs before each session; report and do not use damaged ones • Route and tape cables away from the robot reach and walkways • Only tutor-approved, branded power supplies; no changes to mains equipment • Never open the robot controllers or change their wiring; only lab staff do this	Ongoing	L
[If the PLC extension is attempted] Using the 24 V DC PLC, sensors, safety light curtain and physical e-stop with the robots	Robot restarting unexpectedly after a stop, sensor reset or PLC restart Wiring mistakes on the 24 V PLC, sensors or e-stop Sensors missing a part or giving a false signal (e.g. inductive sensors only detect metal) PLC keeping its old state (outputs, counters, sequence step) after an e-stop or power loss Light curtain or e-stop not working or bypassed (the teaching PLC is not a certified safety system)	Impact injury from an unexpected robot start Minor shock or burn from a wiring fault Robot acting on wrong sensor data or old state False sense of safety leading to injury	Physical e-stop on the workcell Supervised teaching workcell only	• Power isolated while wiring; tutor checks wiring before power is switched on • Choose sensors that suit the parts and test each one before use; the sequence stops if a sensor signal is missing or unexpected • After an e-stop, light-curtain trip or power loss the PLC clears its outputs and sequence state and returns to a known start step – no auto-restart • Restart = check the cell and parts, reset/acknowledge, then a separate start • Test the light curtain, sensors and e-stop at the start of every session and never bypass them • The PLC/e-stop is not described or relied on as a certified safety PLC	Before PLC is first used	L
Manual handling – setting up grippers, fixtures and objects on the bench	Lifting grippers, fixtures or equipment awkwardly Over-reaching across the bench	Back or shoulder strain Equipment dropped on feet	Manual handling training Closed-in shoes	• Two people or a trolley for heavier items; the UR3e stays fixed and is not moved • Use correct lifting technique; ask for help	Ongoing	L
Leaving the workcell unattended between tests or sessions	Robots left powered or with a program running Other people walking into the cell	Impact injury to an unaware person	Supervision during sessions	• Never leave robots enabled or programs running unattended • Stop programs, return robots home, power down robots and PLC, and disconnect Python before leaving	Ongoing	L

Operating while tired, distracted or rushing before deadlines	Reduced attention and reaction time Skipping simulation or safety checks to save time	Operator error leading to collision or injury	Regular breaks Supervision	• Do not operate when tired; work in pairs (operator + spotter) • Plan real-robot testing across drop-in sessions, not the last minute • Stop and ask the tutor if unsure	Ongoing	L
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Approval of assessment	I am satisfied that the residual risk with existing controls is acceptable ☒ Yes ☐ No	Signature	Rudra Narayan Sahu Jayden Kim	Date	30/09/2026
	OR I am satisfied that that the proposed controls will reduce risk to an acceptable level. ☒ Yes ☐ No				

Guidance notes for documenting General Risk Assessments

ACTIVITY

Briefly describe this hazardous work activity - E.g. Operating, Handling, Using ... (Include names) of hazardous equipment, substances or materials used, and any quantities and concentrations of substance(s) or reaction products.

ASSOCIATED HAZARDS

Plant & Equipment – noise, vibration, moving parts (crushing, friction, stab, cut, shear), pressure vessels, lifts/hoists/cranes, sharps

Manual Handling – repetitive movements, lifting awkwardly, lifting heavy objects

Work Environment – moving objects, extremes in temperature, isolation, work at height, allergies to animal bedding, dander and fluids, risk of fire/explosion, slippery surfaces/trip hazards

People – potentially violent or volatile clients/interviewees

Communicable Diseases – exposure to bodily fluids/infectious materials, animal bites and scratches,

Environmental – emissions to atmosphere, discharge to soil and water bodies (including stormwater run-off), nuisance noise & odour, poor ventilation/air quality

Radiation (non-ionizing) – including lasers, microwaves or UV light

Electrical – plug-in equipment used in 'hostile' work environment, exposed conductors, high voltage equipment

Pathogens – dealings with pathogenic microorganisms such as bacteria, parasites, fungi or viruses

GMOs – dealings with genetically modified organisms

Cytotoxins – carcinogens, mutagens or teratogens

Radiation (ionizing) – Ionizing radiation source such as radioactive substance or radionuclide, or irradiating apparatus

Chemical – hazardous substances, dangerous goods, fumes, dust, compressed gas, hazardous waste

INHERENT RISK

Provide details of the harm that could be caused to people or the environment if something goes wrong.

For example: inhalation of fumes, laceration, injury to back, infection, burns to skin or eyes.

Think about what could happen if controls fail or are not in place.

CONTROL MEASURES

Note the existing and proposed actions to reduce risk to an acceptable level. Apply the "Hierarchy of Controls", listed below, when deciding the best control measure to apply. Control types closer the top of the list are preferable.

1. ELIMINATE THE HAZARD. For example: use a different less dangerous piece of equipment, fix faulty machinery, use safer materials or chemicals
2. ISOLATE THE HAZARD FROM THE PEOPLE. Separate people from the danger. For example: use shielding, use lifting equipment or trolleys, remove dust or fumes with exhaust system, lock-out machinery.
3. CHANGE THE WAY THE JOB IS DONE. For example: change work practices, provide training, information and signs, develop work procedures.
4. USE PERSONAL PROTECTIVE EQUIPMENT (PPE), noting specific PPE is required for each job. For example: respirator, hearing protection, gloves. Training and information is required for the use of PPE.

RESIDUAL RISK LEVEL (H, M, L)

Estimate risk taking into account the way the activity is run and control measures put in place. The level of risk can be determined by combining consequence and likelihood using the risk matrix from below. Residual risk should be reduced to a level acceptable by management.

CONSEQUENCE OF HARM - This is how bad it will be if something does go wrong e.g. the number of people that could be harmed, the severity of injury.

LIKELIHOOD OF HARM - Chance of harm occurring is affected by the duration of the activity and its frequency; the number of people doing the activity and the level of exposure to the hazard.

		CONSEQUENCE					
		Insignificant	Minor	Moderate	Major	Catastrophic	
		Injury/illness consequence	Non-injury incident	Injury/ill health requiring first aid	Injury/ill health medical attention	Injury/ill health requiring hospital admission	Fatality or permanent disabling injury
		Environmental consequence	Minor effects on biological or physical environment	Moderate short term effects but not effecting ecosystem functions	Serious medium-term environmental effects	Very serious long term impairment of ecosystem functions	
LIKELIHOOD	Almost Certain	The event will occur on an annual basis	Moderate	High	High	Critical	Critical
	Likely	The event has occurred several times or more in your career	Moderate	Moderate	High	High	Critical
	Possible	The event might occur once in your career	Low	Moderate	Moderate	High	High
	Unlikely	The event does occur somewhere from time to time	Low	Low	Moderate	Moderate	High
	Rare	Heard of something like this occurring somewhere	Low	Low	Low	Moderate	Moderate

UTS SAFE WORK METHOD STATEMENT (SWMS)

1. FACULTY/SUBJECT			
Faculty/Subject title		41013 Industrial Robotics	
Subject supervisor/coordinator		Gavin Paul	
SWMS prepared by		Rudra Narayan Sahu, Jayden Kim	

2. WORK ACTIVITY DESCRIPTION	
Describe the work activity E.g. Operating, Handling, Using.. Include names of hazardous equipment, substances or materials used, and any quantities and concentrations of substance(s) or reaction products.	Group pick-and-place workcell for Lab Assignment 2 (DoGoodBot) in CB11.10.403. A real UR3e cobot is controlled from Python via ROS (roslibpy through the Raspberry Pi) using the group's GUI with jog/teach control and a software e-stop. It works alongside a Stäubli TX2-60: a 6-axis industrial robot (up to 4.5 kg payload) with a CS9 controller, SP2 teach pendant and SIL3-PLC safety functions (Safe Speed, Safe Zone, Safe Stop). The Stäubli is simulated in this project but is assessed here as if installed in the real workcell. Because it is faster and stronger than the UR3e, it would only be run inside a guarded cell, in manual/reduced speed, under supervision. The robots pick and place light, blunt test objects (plastic/foam blocks) well below their rated payloads. Most of the system is developed and tested in the Swift simulation first; real hardware is only run under tutor supervision, at reduced speed, inside a marked keep-out zone. Optional extension (only if attempted): a 24 V DC PLC coordinating photoelectric/capacitive/inductive/limit sensors, a safety light curtain and a physical e-stop. No hazardous chemicals or substances are used.

3. HAZARDS: Choose those hazard types that will need to have control measures in Section 4			
Work Environment - Working in Remote Locations - Working Outdoors/fieldwork - Clinical/Industrial setting - Poor ventilation/Air quality - Temperature extremes - Working at Height - Slip/Trip/Fall hazards	Plant - Noise - Vibration - Working with compressed air - Lifts Hoists or Cranes - Moving parts (Crushing, friction, cut, stab, shear hazards) - Pressure Vessels or Boilers	Chemical - Hazardous Chemicals use - Skin/eye irritant - Sensitiser - Mutagen - Carcinogen - Toxic to reproduction - Aquatic toxicity - Toxic - Corrosive - Dangerous when wet	Ergonomic/Manual Handling - Repetitive or awkward movements - Lifting heavy objects - Over reaching - Working above shoulder or below knee height - Poor workstation set up
Electrical - Plug in equipment - High voltage - Exposed wiring - Exposed conductors	Radiation - Ionising Radiation - Non-ionising radiation (Lasers, Microwaves, Ultraviolet light)	Biological - Sharps/Needles - Cytotoxins - Pathogens/infectious materials - Infectious materials - Communicable diseases - Animal/insects - Work with fungi/bact/viruses	Psychosocial - Aggressive or violent clients/students - Working in isolation - Working with timeframes - Staffing issues

4. CONTROLS MEASURES: Choose those that apply for hazards identified		
Eliminate/Isolate/Substitute / Engineering Controls - Remove hazard - Restrict access - Redesign equipment - Guarding / Barriers / Fume Cupboard / exhaust - Biosafety cabinet - Use safer materials/substances - Ventilation - Regular maintenance of equipment - Redesign of workspace / workflow	Admin specific: Licenses/permits Work Methods - Training Information or Instruction - Licensing or certification of operators - Test and tag electrical equipment - Restricted access - Regular breaks - Task rotation - Work in pairs - Document Chemical risk assessment - Ladder / Sling register	Emergency Response Systems - First aid kit - Chemical spill kit - Safety shower - Eye wash station - Emergency Stop button - Remote Communication Mechanism

Other controls not listed

Every motion is simulated in Swift first. Real robots run at reduced speed with conservative targets, with one command source at a time and a spotter at the e-stop.

Marked keep-out zone around the robots; the robots do not move in a shared zone at the same time. People are warned before any motion.

The GUI/software e-stop is secondary to the hardware e-stop. After any stop there is a deliberate reset, then a separate resume, and no automatic restart.

If the PLC is used: wiring is checked by the tutor before power-up. The light curtain and e-stop are tested every session and never bypassed. The PLC is not treated as a certified safety PLC.

5. PPE REQUIRED (Tick those that apply)



EYE PROTECTION



HEARING PROTECTION



CLOSED-IN FOOTWEAR MUST BE WORN



HAND PROTECTION



FACE SHIELD



PROTECTIVE CLOTHING



RESPIRATORY PROTECTION



LONG HAIR MUST BE CONTAINED



HEAD PROTECTION



6. EMERGENCY EQUIPMENT



EMERGENCY SHOWER



EMERGENCY SPILL KIT INSIDE



EMERGENCY EYEWASH



7. WORK ACTIVITY STEPS

BEFORE YOU START:

- All group members have completed the CB11.10.403 lab induction and read this RA/SWMS; a tutor is present
- Agree roles: operator (runs code), spotter (hand at e-stop); agree which robot(s) will move
- Check the Stäubli cell/guarding is in place and its Safe Zone/Safe Speed settings are active
- Inspect the UR3e, Stäubli TX2-60, grippers, cables (and sensors, light curtain, PLC wiring if used)
- Clear the bench, mark the keep-out zone and use only light, blunt test objects
- Locate and test the UR3e pendant e-stop, the Stäubli SP2 pendant e-stop and enabling device (and the PLC physical e-stop)
- Run the full sequence in the Swift simulation and check frames, joint limits and collisions
- Set the UR3e to reduced speed (speed slider $\leq 30\%$)

STEPS IN WORK ACTIVITY:

1. Power on the UR3e from the teach pendant, release brakes and start the ROS listener; power on the Stäubli CS9, select manual mode at reduced speed and home it with everyone outside its cell
2. Connect the laptop to ROS (roslibpy) and check the live joint states match the simulation
3. Announce the motion; everyone except the operator and spotter stays outside the keep-out zone
4. Test each motion segment on its own at reduced speed – one robot at a time
5. Test the GUI e-stop and hardware e-stop, then the reset and separate resume
6. If the PLC is used: power on 24 V only after the tutor wiring check; check each sensor detects the parts correctly, and check that a light-curtain trip or e-stop stops everything and the PLC returns to its start step
7. Run the full integrated pick-and-place at reduced speed; only increase speed with tutor approval
8. When finished: stop programs, home the robots (Stäubli back to its home pose), power down robots and PLC, disconnect and tidy cables

EMERGENCY PROCEDURES:

- Press the nearest e-stop (UR3e pendant, Stäubli SP2 pendant, PLC physical e-stop)
- Keep everyone clear and tell the tutor; do not reset until the cause is found

- Recover: reset/acknowledge, clear queued robot commands and PLC sequence state, check parts in the cell, re-read joint states, return home at low speed, then a separate start
- For injuries use the first aid kit, notify the tutor, and notify security or dial 6 on the UTS internal phone; report the incident

TRAINING REQUIRED:

- CB11.10.403 Mechatronics lab induction (Rapid Global)
- Tutor instruction on the UR3e teach pendant, Python/ROS control, e-stop and recovery
- Tutor/lab-staff instruction on the Stäubli TX2-60, CS9 controller and SP2 pendant (and the PLC workcell) before first use
- All group members have read and signed this RA and SWMS

8. SIGN OFF

PREPARED BY:

LAB SUPERVISOR

DATE: 30/09/2026NAME: RUDRA NARAYAN SAHU,
JAYDEN KIMNAME: FELIPE ULLOA GUTIERREZREVIEW DATE: 30/09/2026